

PopNavShift: Stress-Testing Social Navigation under Behavioral Population Shift

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Project webpage: tantansir.github.io/PopNavShift

Abstract—Social-navigation algorithms are often evaluated under a fixed pedestrian-behavior distribution, despite substantial variation in pedestrian responses to robots across individuals and social contexts. We introduce PopNavShift, a matched simulation framework for stress-testing social-navigation strategies under pedestrian population shifts. PopNavShift constructs population-conditioned pedestrian motion profiles by prompting Gemini 3.7 Flash with 600 synthetic persona records from MatrAIx Persona 1M and deterministically mapping the responses into bounded motion parameters. It then compares three representative navigation strategies, reactive avoidance, early yielding, and reciprocal collision avoidance, across eight population conditions and 7,488 matched robot runs. In a matched intervention on the same 202 personas, changing only time pressure reverses 8.6% of controller rankings based on robot travel time, but 22.4% based on mean pedestrian delay and 23.9% based on worst-decile delay. Across population conditions, this sensitivity is greater for pedestrian burden than for robot travel time and increases in spatially constrained settings; the same qualitative pattern persists under a second pedestrian dynamics model. These findings support evaluating navigation strategies across behavioral populations using both robot performance and pedestrian burden.

I. INTRODUCTION

As robots increasingly operate in human-shared spaces, their navigation must account not only for task efficiency but also for how nearby pedestrians respond to them. These responses can vary substantially across people and situations, with differences in how much clearance pedestrians maintain, when they yield, whether they accept small gaps, and how quickly they move around an approaching robot. Human-robot proxemics and social-navigation studies show that these behaviors depend on robot motion, orientation, gaze, encounter context, and social conventions [1]–[4]. In simulation-based evaluation, pedestrian behavior is often represented through a selected model or behavioral distribution, making the represented population part of the evaluation setup alongside the physical scenario.

Existing work accounts for variation in pedestrian behavior. Social-navigation methods incorporate pedestrian emotion, individual preferences, and social judgments into planning [5]–[7], while platforms such as SOCIALGYM and models such as MAC-ID support configurable or heterogeneous pedestrian behaviors [8], [9]. However, these approaches do not directly test whether the relative performance of the same navigation strategies remains stable when the represented

pedestrian population changes. Navigation strategies can distribute interaction adjustment differently between the robot and nearby pedestrians. Changes in pedestrian yielding, clearance preference, or gap acceptance may therefore alter their relative performance even when geometry, density, and initial conditions remain fixed. Although current evaluation guidelines emphasize human, environmental, and task context [10], the stability of navigation-strategy rankings across behavioral population shifts has received limited direct study.

We therefore introduce **PopNavShift**, a matched evaluation framework that holds physical episodes and navigation strategies fixed while varying the represented pedestrian population (Fig. 1). This work makes three contributions. First, we establish behavioral population shift as an explicit dimension of social-navigation evaluation, making robustness to population change a direct evaluation target. Second, we operationalize population robustness through matched population replay and a same-persona time-pressure intervention, assessing changes in navigation-strategy rankings based on robot travel time and pedestrian burden, including mean and worst-decile added delay relative to no-robot baselines. Third, we examine how population sensitivity varies with spatial constraints and whether ranking sensitivity persists across alternative response-to-motion mappings, pedestrian dynamics models, and language models.

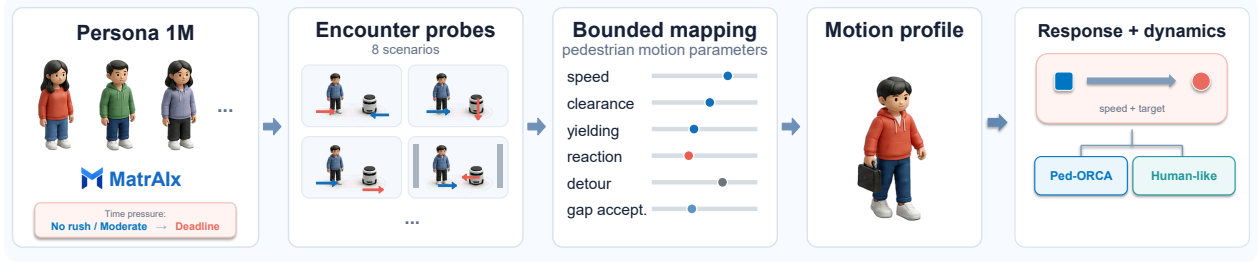
II. RELATED WORK

A. Human Variation in Social Navigation

Human-robot interaction studies document variation in proxemics, perceived safety, passing conventions, and preferred paths in shared environments [4], [10]. Robot approach direction, orientation, gaze, and motion can affect the spacing people maintain during encounters [1]–[3]. These findings show that pedestrian spatial behavior depends on encounter geometry and interaction cues.

Behavior also varies across populations and situational contexts. Eresha et al. report cultural differences in proxemic behavior around humanoid robots [11], while Xiao et al. compare collision-avoidance behavior among participants in the United States and Israel [12]. Time pressure can affect crossing speed and gap acceptance [13], motivating its use as the controlled contextual variable in our matched intervention.

a From persona records to pedestrian motion



b Matched replay of one encounter

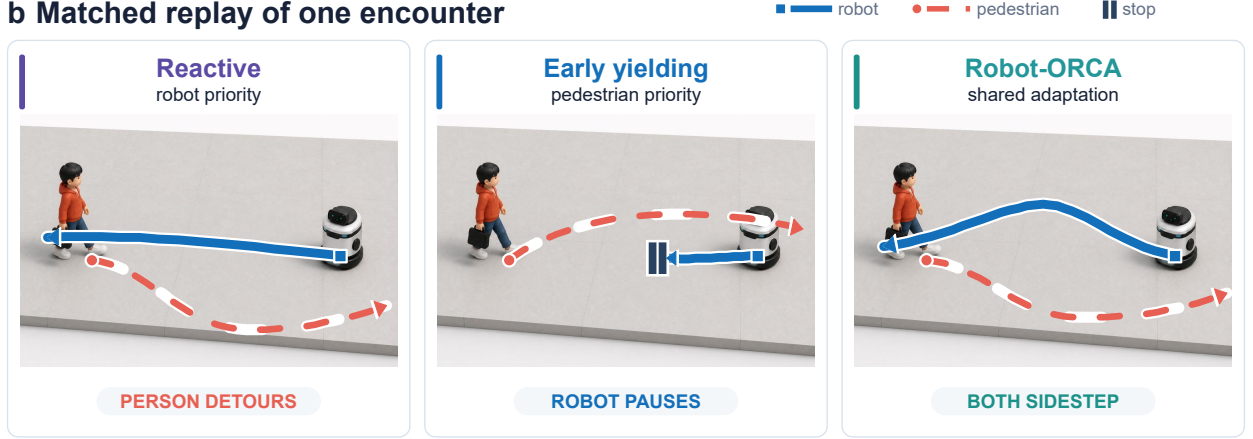


Fig. 1. PopNavShift evaluates social navigation strategies under behavioral population shift. (a) Persona records are converted through standardized robot encounter probes and a deterministic bounded mapping into pedestrian motion profiles. The matched intervention changes time pressure for the same persona records before repeating the probe-to-motion pipeline. (b) Identical physical encounters are replayed under reactive geometric, early yielding, and Robot-ORCA controllers. Ped-ORCA is the primary pedestrian dynamics backend; navground Human-like behavior is used for robustness analysis.

Other work incorporates individual states and preferences into robot navigation. EWareNet uses inferred pedestrian emotion to adapt intent prediction and spatial profiles [5]; Choi et al. infer user-specific navigation preferences [6]; and NaviDIFF incorporates human preferences and social principles into navigation behavior [7]. These methods adapt robot behavior to individual users, preferences, or inferred states. PopNavShift keeps the navigation strategies fixed and examines whether their relative rankings remain stable as the represented pedestrian population changes.

B. Pedestrian Diversity in Simulation

To represent variation in pedestrian behavior, social-navigation simulation relies on motion models that specify how people move and interact around robots. Classical approaches such as the Social Force model and reciprocal collision avoidance generate pedestrian motion from explicit interaction rules [14], [15], while other approaches learn crowd motion or interaction behavior from observed trajectories. These models provide the basis for simulating pedestrian responses, with behavioral variation typically represented through shared model parameters or predefined behavior settings.

Recent simulation systems make this variation more explicit. SOCIALGYM exposes configurable human parameters for social-navigation experiments [8], while SEAN 2.0 supports multiple methods for specifying pedestrian motion and

generating varied social situations [16]. MAC-ID models individual differences through a local coordination factor that varies pedestrian behavior from more competitive to more cooperative interaction, and its BSON benchmark uses these behaviors to construct diverse social environments for navigation evaluation [9]. These systems show that pedestrian behavior can be varied systematically within simulation, providing a basis for evaluating navigation methods across different behavioral conditions.

Recorded human-motion datasets provide another source for grounding simulated pedestrian behavior. SCAND, THÖR, and THÖR-MAGNI capture pedestrian motion and human-robot encounters in real environments [17]–[19]. Such datasets can inform plausible motion patterns and physical parameter ranges. Our matched population-shift design additionally requires structured contextual attributes that can be modified while keeping the underlying persona fixed. The recorded datasets above do not provide this intervention structure.

C. Population Robustness in Social Navigation

Alongside work on modeling pedestrian diversity, prior research has developed standardized protocols and benchmarks for evaluating social-navigation methods. SocNavBench replays recorded pedestrian trajectories under a common evaluation protocol [20], while SocNav1 provides socially judged navigation situations for evaluating social conventions [21]. Recent evaluation guidelines further organize metrics,

scenarios, and human, environmental, and task contexts for social-navigation assessment [10].

BSON is especially relevant because it evaluates navigation methods in environments with diverse pedestrian interaction styles [9]. Built on MAC-ID, it uses pedestrian coordination styles ranging from more competitive to more cooperative behavior to construct diverse social environments. PopNavShift focuses on whether the relative performance of the same navigation strategies remains stable as the represented pedestrian population changes while the physical encounter is held fixed. We evaluate this through matched population replay, comparing changes in controller rankings and pedestrian burden across population conditions.

III. METHOD

A. Framework Overview

To operationalize the matched population-replay design, PopNavShift separates the construction of behavioral populations from pedestrian dynamics and robot control (Fig. 1). Eight fixed robot-encounter probes, each describing a standardized navigation situation, elicit structured responses from each persona, which a deterministic mapping converts into seven bounded motion controls. The resulting motion profiles are then organized into eight population conditions for matched evaluation using a shared pedestrian response layer and continuous dynamics model.

B. Persona Records and Standardized Probes

We sample 600 records from the public MatraIx Persona 1M release [22], [23]. The prompt includes available fields related to time pressure, safety and risk preferences, decision style, trust in technology, patience, accessibility needs, and mobility. Demographic attributes are excluded from the prompt and mapping.

Gemini 3.7 Flash receives each record and answers eight fixed robot encounter probes. For each encounter, it selects one action from *continue*, *slow*, *stop*, *yield*, *pass*, or *detour*. The probes vary relative approach direction, available space, crowding or obstruction, and robot behavior. Together, they cover situations in which clearance, yielding, speed adjustment, and detouring can affect the interaction.

After selecting the encounter actions, the model returns an ordered response block covering preferred clearance, willingness to wait, yielding, small-gap acceptance, detouring, robot trust, and urgency on five-level scales. The complete structured response is cached for each persona and reused across simulation scenarios. The probes form a response-elicitation battery; they are separate from the physical scenarios used in the simulation experiments.

C. Mapping from Responses to Motion

The probe responses are converted into continuous pedestrian controls using a deterministic bounded mapping. The five ordered response levels are encoded as $\{0, 0.25, 0.5, 0.75, 1\}$. For persona i , the motion profile is

$$\theta_i = (v_i, d_i, a_i, y_i, \tau_i, L_i, g_i),$$

TABLE I
BEHAVIORAL POPULATION CONDITIONS AND COHORT SIZES

Condition	Construction	n
Homogeneous	Overall median parameter vector	1
Mixed	All mapped records	600
High time pressure	Deadline or Emergency source field	49
Mobility support	Explicit mobility or motor-support field	45
High caution	Upper quartile of clearance/yield/gap score	170
Low robot trust	Low or very-low probe response	197
Matched base	Original no-rush/moderate records	202
Matched deadline	Same IDs; time pressure set to Deadline	202

where the components denote preferred speed, robot clearance, avoidance strength, yielding propensity, reaction delay, detour threshold, and gap acceptance, respectively. These are operational simulator controls rather than estimated psychological traits.

Let q_i^c , q_i^w , q_i^y , q_i^g , q_i^d , q_i^t , and q_i^u denote the normalized responses for clearance, waiting, yielding, small-gap acceptance, detouring, robot trust, and urgency. Mapping A also summarizes the categorical encounter actions using fixed ordered scores, whose mean is q_i^a . Its caution score is

$$C_i = 0.34q_i^c + 0.22(1 - q_i^g) + 0.22q_i^y + 0.22q_i^a. \quad (1)$$

The normalized controls combine these responses according to their intended simulation roles: urgency affects preferred speed and yielding; clearance and gap acceptance use their corresponding responses directly; avoidance combines caution with robot trust; reaction delay combines urgency and trust; and detour tendency combines the reported detour response with caution. All coefficients are fixed before simulation and are not fitted to observed trajectories.

Each normalized control z_i^k is clipped to $[0, 1]$ and mapped to its physical range $[\ell_k, h_k]$ by

$$\theta_i^k = \ell_k + z_i^k(h_k - \ell_k). \quad (2)$$

The physical ranges are 0.60–1.80 m/s for preferred speed, 0.30–1.20 m for robot clearance, 0.60–2.20 for avoidance strength, 0.15–1.20 s for reaction delay, and 1.0–4.0 m for the detour threshold. Yielding and gap acceptance remain in $[0, 1]$. Mobility-related fields apply ordered speed caps after mapping. The resulting speed and spacing ranges overlap values reported in THÖR and related datasets [18].

We additionally test three alternative mappings from responses to motion. Mappings B and C modify the weighting and action-response construction, while Mapping D uses separable response dimensions without weighted mixtures. All four mappings retain the same physical ranges and mobility caps. Mapping A is used in the main experiment and Mappings B–D are used for sensitivity analysis.

D. Population Conditions and Matched Intervention

Table I defines the eight population conditions used in the experiments. These selected behavioral contrasts can overlap in persona membership. Figure 2(a) summarizes the resulting distributions over the motion parameters.

The matched time-pressure intervention provides the main controlled population shift. It contains the same 202 persona

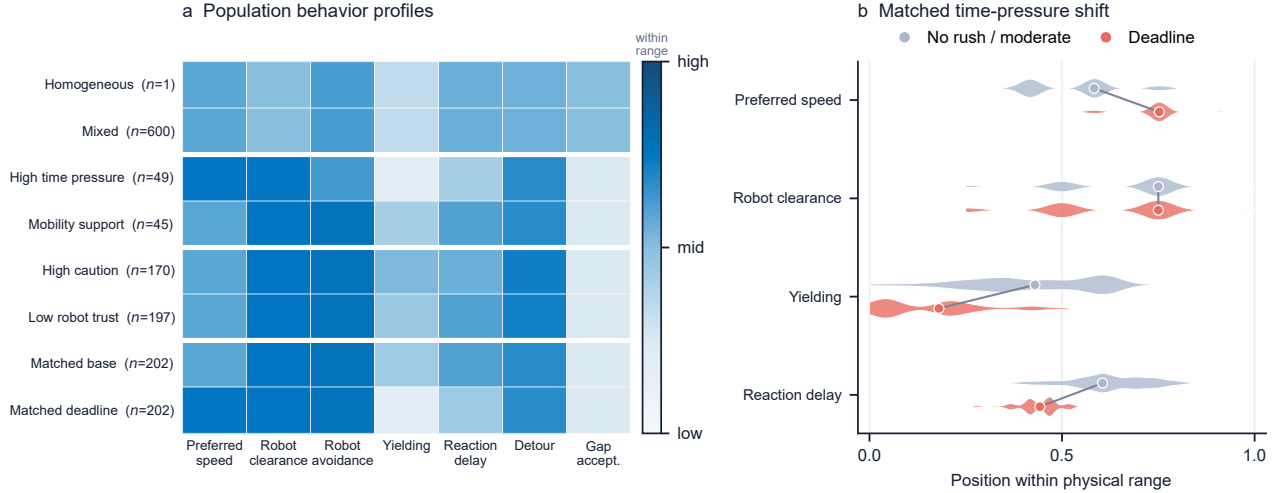


Fig. 2. Pedestrian motion profiles across population conditions. (a) Median position within each parameter’s physical range; labels report cohort size. White separators distinguish reference conditions, source-field cohorts, probe-defined cohorts, and the matched pair. (b) Profile distributions for the same personas before and after the matched change in time pressure from *No rush/Moderate* to *Deadline*.

records in both conditions: 113 originally have *No rush* and 89 have *Moderate* time pressure. The matched-base condition uses the original records. For the matched-deadline condition, only the time-pressure field is changed to *Deadline*, after which the same encounter probes and mapping from responses to motion are rerun. Persona identity and all other source fields remain unchanged.

Time pressure is used as a mutable contextual variable because controlled pedestrian studies have found that imposed time pressure can change crossing speed and gap acceptance [13]. This evidence motivates the intervention; the observed effect sizes are not used to calibrate the persona responses or motion parameters. Figure 2(b) shows how the resulting motion-profile distributions change for the same personas.

For episodes containing multiple pedestrians, profiles are sampled with replacement using fixed episode-specific assignments. Matched-base and matched-deadline runs use the same persona at every pedestrian slot. Any difference between the two conditions therefore originates from responses changed by the time-pressure intervention rather than from different sampled identities.

E. Pedestrian Response and Dynamics

Each motion profile enters simulation through a shared robot-conditioned pedestrian response layer. The response activates after the robot enters sensing range and the profile-specific reaction delay has elapsed. Both agents are projected over a four-second horizon using the effective clearance

$$d_i^{\text{eff}} = d_i(1 - 0.35g_i).$$

A conflict is predicted when the closest projected center distance falls below the sum of the two agent radii and d_i^{eff} .

During a predicted conflict, yielding propensity reduces the pedestrian’s preferred speed as the robot approaches the detour threshold L_i . When the robot is nearly stationary, this reduction is relaxed to allow the pedestrian to pass. Inside L_i , the response layer places a lateral waypoint 1.5 m ahead,

with offset determined by effective clearance and avoidance strength. Each episode fixes a passing side and switches sides if an obstacle blocks the preferred route.

Navground provides the continuous two-dimensional simulation [24]. The primary pedestrian backend, Ped-ORCA, uses omnidirectional Optimal Reciprocal Collision Avoidance (ORCA) with a 4.5 m spatial horizon [15]. A robustness experiment uses navground’s Human-like (HL) behavior, which responds to visual collision cues within a 5 m horizon [25]. Both backends receive the same preferred speeds and targets from the shared response layer. The simulation step is 0.1 s and the episode time limit is 45 s.

F. Robot Controllers

The robot is modeled as a differential-drive disk with radius 0.35 m. All controllers share the same goal, footprint, dynamic limits, local observations, and collision geometry.

The *reactive geometric* controller samples feasible wheel commands over a 2 s horizon with a 0.04 m safety buffer and adjusts its trajectory when a conflict becomes imminent. The *early yielding* controller detects conflicts over a 4.5 s horizon, moves toward a clear pull-aside pose with a 0.16 m buffer, and waits while the pedestrian crosses the conflict region. The *Robot-ORCA* controller uses reciprocal collision avoidance with a 5 m spatial horizon and a 0.12 m safety margin under navground’s two-wheel kinematics.

The controllers therefore differ primarily in how interaction adjustment is allocated. Reactive control postpones robot-side adjustment, early yielding gives greater priority to the pedestrian, and Robot-ORCA uses reciprocal adaptation. Ped-ORCA denotes the pedestrian dynamics backend; Robot-ORCA denotes the reciprocal robot controller.

IV. EXPERIMENTAL DESIGN

A. Matched Evaluation Grid

We evaluate PopNavShift on five types of physical encounters: a bidirectional sidewalk, a head-on encounter, a crossing,

a bottleneck, and a static obstruction. The scenario grid varies available width (1.8, 2.4, 3.0, and 4.2 m), pedestrian density (0.06, 0.12, and 0.20 persons/m²), and robot speed (0.60 and 1.00 m/s). Head-on encounters contain one pedestrian and use the lowest density, yielding 104 valid physical cells rather than a full Cartesian product.

Each physical cell contains three fixed variants with different pedestrian start positions and walking directions, giving 312 episodes in total. Every episode is replayed with all eight population conditions and all three robot controllers. For each population–episode combination, an additional run without a robot provides the pedestrian baseline. The matched design keeps physical geometry, episode initial conditions, and controller definitions unchanged when the represented pedestrian population changes.

B. Metrics and Ranking Stability

Robot-centered outcomes include success, travel time, path length, stopping time, pedestrian and world collisions, and minimum pedestrian clearance. Pedestrian-centered outcomes include travel time, walking distance, stopping time, detour distance, and minimum robot clearance.

Our main pedestrian-burden measure is added travel time. For pedestrian j ,

$$\Delta T_j = T_j^{\text{robot}} - T_j^{\text{no robot}},$$

where the run without a robot uses the same physical episode and pedestrian assignment. Added distance is defined analogously. A paired delay is available when the pedestrian completes both runs.

A physical cell fixes scenario type, width, density, and robot speed and pools its three episode variants. Mean pedestrian delay is computed over all available paired records in the cell. To capture concentrated burden, worst-decile delay is the mean of the largest $\lceil 0.1n \rceil$ paired delays. Failed or timed-out robot runs retain the 45 s episode limit as robot travel time.

For each physical cell and metric, we compare every pair of controllers separately under each population. A *ranking reversal* occurs when a non-tied controller ordering under one population changes sign under another. We use tie tolerances of 0.02 s for pedestrian delay and 0.05 s for robot travel time. We additionally report a completed-cell analysis restricted to cells in which all three controllers finish every episode variant under both populations being compared.

To quantify the magnitude of population effects independently of controller ordering, we define *population sensitivity* as the range of a metric across population conditions for a fixed controller and physical cell.

C. Sensitivity Analyses

We test whether the observed population dependence is specific to the mapping from responses to motion or pedestrian dynamics model. Mapping sensitivity is evaluated on twelve representative episodes spanning narrow and wide sidewalks, bottlenecks, and static obstructions. Mappings A–D use the same cached responses, population membership, and episode assignments. In addition to ranking-reversal rates, we measure

agreement with Mapping A on eligible non-tied controller comparisons.

We also perturb the structure of the pedestrian profiles. One analysis reassigns complete motion-profile vectors across persona identities before reconstructing five shared populations. A second independently shuffles each motion-parameter column within a population, preserving marginal parameter distributions while breaking their cross-parameter association. Separate runs isolate the contributions of time-pressure, caution, and trust response blocks.

Finally, we repeat a subset of the evaluation with nav-ground’s Human-like pedestrian dynamics. This comparison uses twelve obstacle-free sidewalk, head-on, and crossing episodes and four shared populations: homogeneous, mixed, matched base, and matched deadline. Physical scenes, persona assignments, response rules, and robot controllers remain matched between Ped-ORCA and Human-like dynamics. Backend comparisons use the intersection of cells completed by all three controllers under both models, after which the same metric-specific tie tolerances are applied separately to each backend.

V. RESULTS

A. Matched Time-Pressure Shift

The main experiment contains 7,488 robot runs and 2,496 matched runs without a robot, with paired pedestrian delay available for nearly all records. Changing time pressure from *No rush/Moderate* to *Deadline* alters at least one categorical response for 201 of the 202 matched personas.

This shift changes controller comparisons more often for pedestrian burden than for robot travel time. Mean-delay orderings reverse in 22.4% of comparable cells and worst-decile-delay orderings in 23.9%, compared with 8.6% for robot travel time. The effect also differs across controllers. As shown in Fig. 3, the matched deadline shift reduces pedestrian delay under reactive control and Robot-ORCA but increases it under early yielding. The population change therefore alters the relative burden associated with different navigation strategies rather than producing a common shift across controllers.

B. Comparisons across Population Pairs

The same pattern appears across all 28 population pairs. Table II pools the three controller pairs after excluding ties. Robot travel-time orderings reverse in 8.7% of eligible comparisons, compared with 25.8% for mean pedestrian delay and 24.8% for worst-decile delay.

Restricting the analysis to cells in which all three controllers complete every episode under both populations reduces the reversal rates to 5.0%, 21.5%, and 22.1%, respectively. The higher sensitivity of pedestrian-burden rankings therefore remains after controller failures are removed from the comparison.

C. Aggregate Controller Outcomes

The controllers also exhibit distinct aggregate performance trade-offs. Table III summarizes outcomes over the full

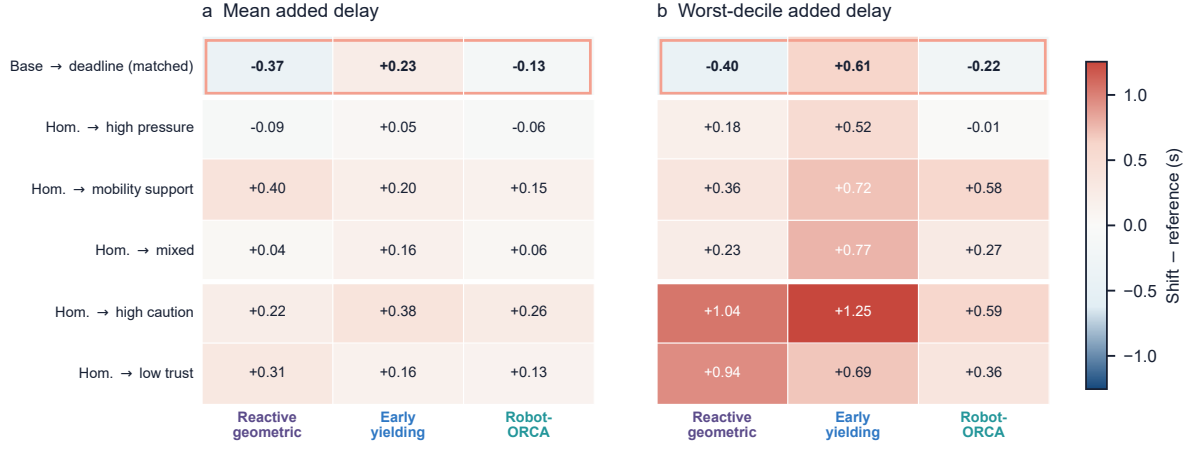


Fig. 3. Changes in pedestrian delay under behavioral population shift. Entries show shifted minus reference delay over the same 104 physical cells. The outlined row shows the matched time-pressure intervention on the same personas. White separators distinguish the matched intervention, source-field cohorts, the mixed population, and probe-defined cohorts.

matched grid. Reactive control achieves the highest completion rate at 90.3%, while Robot-ORCA produces the lowest mean pedestrian delay (0.65 s) and worst-decile delay (2.80 s). Early yielding has the lowest mean delay among the two non-reciprocal controllers but also the highest timeout rate.

Robot-ORCA performs best on aggregate pedestrian delay, whereas reactive control completes more runs. Most Robot-ORCA collisions are contacts with the environment rather than pedestrians, accounting for the difference between its pedestrian- and world-collision rates in Table III.

D. Spatial Constraint and Tail Burden

Pedestrian burden is concentrated in the tail. Across the controllers, worst-decile pedestrian delay is approximately four times the corresponding population mean. Population shifts can also affect the tail differently from the mean, as illustrated by the early yielding response to the matched time-pressure intervention in Fig. 3.

Available space strongly affects the magnitude of population sensitivity. As shown in Fig. 4, the range of mean pedestrian delay across populations is larger on narrow sidewalks and bottlenecks. Pooled population sensitivity decreases from 2.23 s at 1.8 m width to 0.50 s at 4.2 m, with the same overall direction for each controller. When analysis is restricted to cells completed by all controllers, width is no longer monotonic, indicating that part of the stronger narrow-space effect coincides with controller failures.

E. Mapping and Profile Structure

The population effect persists when the mapping from responses to motion is changed. Table IV compares the four

TABLE III

CONTROLLER OUTCOMES OVER THE FULL MATCHED GRID

Outcome	Reactive	Early yielding	Robot-ORCA
Success (%)	90.3	81.5	87.8
Pedestrian collision (%)	2.6	5.4	2.1
World collision (%)	0.1	0.0	7.1
Timeout (%)	7.0	13.0	3.1
Robot travel time (s)	22.9	28.4	24.2
Minimum clearance (m)	0.70	0.70	0.73
Mean added delay (s)	1.25	1.06	0.65
Worst-decile delay (s)	5.13	4.42	2.80

TABLE IV

RANKING REVERSALS AND AGREEMENT ACROSS MAPPINGS FROM RESPONSES TO MOTION

Mapping	Reversal rate		Agreement with A	
	Mean	Tail	Mean	Tail
A: original	24.5%	26.0%	—	—
B: ordered	24.2%	28.6%	92.3%	92.6%
C: equal weights	24.6%	25.7%	92.2%	91.8%
D: direct	37.5%	35.8%	76.8%	71.0%

mappings on the representative subset. Mappings B and C preserve more than 91% of Mapping A's mean- and tail-delay controller orderings and produce similar reversal rates. The simpler direct Mapping D yields higher reversal rates, reaching 37.5% for mean delay and 35.8% for worst-decile delay.

Changing the association structure among persona profiles also changes the magnitude of the effect. Across five shared populations, randomly reassigning complete motion profiles among persona identities reduces mean/tail reversal rates from 28.4%/26.9% to 19.8%/18.8%. Independently shuffling each parameter column gives 26.8%/29.7%. Thus, shifted parameter distributions continue to produce ranking changes after profile associations are disrupted, while the associations themselves affect how frequently reversals occur. In the single-block analyses, time pressure and caution produce more

TABLE II

REVERSALS IN CONTROLLER RANKINGS ACROSS POPULATION PAIRS

Metric	All cells	Completed cells
Robot travel time	8.7%	5.0%
Mean pedestrian delay	25.8%	21.5%
Worst-decile delay	24.8%	22.1%

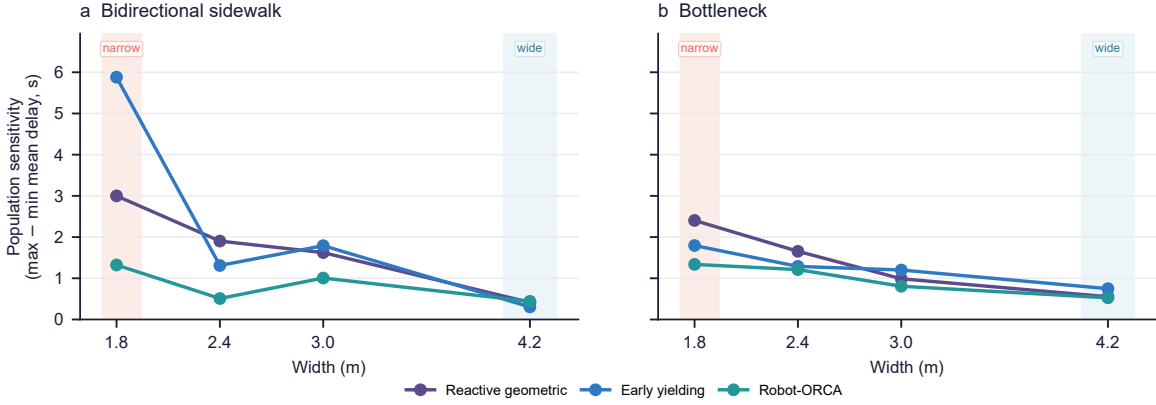


Fig. 4. Population sensitivity across available widths. Population sensitivity is the range of mean pedestrian delay across population conditions for a fixed controller and physical cell, shown for (a) bidirectional sidewalks and (b) bottlenecks.

ordering changes than robot trust alone.

F. Pedestrian-Dynamics and LLM Robustness

The qualitative difference between pedestrian burden and robot travel time also appears under a second pedestrian dynamics model. On cells completed by all three controllers under both backends, Ped-ORCA produces mean- and worst-decile-delay reversal rates of 10.5% and 9.8%, compared with 3.2% for robot travel time. Under the Human-like backend, the corresponding rates are 35.2%, 34.6%, and 2.4%. The two dynamics models differ substantially in reversal magnitude, while both produce more pedestrian-burden ranking reversals than robot travel-time reversals.

Repeating the matched intervention with Gemini 3.5 Flash Lite gives the same qualitative pattern. With the same persona IDs, reversal rates are 25.6% for mean pedestrian delay, 26.5% for worst-decile delay, and 9.1% for robot travel time. Probe-defined cohort membership can change across language models, so this comparison is restricted to the matched intervention.

VI. DISCUSSION

A. Population-Conditioned Evaluation

The observed ranking reversals indicate that comparative evaluations of social-navigation strategies can depend on the behavioral population used for evaluation. With physical episodes and navigation strategies held fixed, changes in pedestrian motion profiles frequently altered strategy rankings based on pedestrian burden. This extends prior work on human variation in social navigation by showing that behavioral diversity can affect not only the simulated interaction itself, but also the comparative evaluation of navigation strategies.

The stronger sensitivity of pedestrian burden suggests that population shift can affect how interaction costs are distributed between the robot and nearby pedestrians, even when robot-centered performance remains comparatively stable. Variation in speed, clearance, yielding, reaction, detouring, and gap acceptance can interact differently with navigation strategies, so a strategy comparison that appears stable in robot travel time may still vary substantially in pedestrian burden across

populations. These patterns suggest population-specific design considerations, including clearer passage negotiation under time pressure, less forced slowing or detouring for mobility-support profiles, and greater usable clearance or earlier, more predictable conflict resolution for cautious or low-trust profiles. Such differences may become more consequential in narrow sidewalks and bottlenecks, where limited spatial flexibility leaves less room to absorb behavioral variation.

B. Implications for Social Navigation Evaluation

The population dependence observed here suggests that behavioral population composition should be treated as an evaluation dimension alongside geometry, density, and other scenario conditions. A benchmark can include heterogeneous pedestrians while still evaluating every navigation strategy under a single fixed mixture of behaviors; such a setup may obscure population-specific changes in relative performance. Matched population replay provides a direct way to assess this dependence by evaluating the same physical episodes across alternative pedestrian populations and reporting robot-centered outcomes together with mean and tail pedestrian burden. Population robustness therefore provides an additional dimension for social-navigation evaluation.

To extend population-robustness evaluation beyond controlled synthetic populations, the same framework could be extended to empirically measured populations. Future datasets that combine contextual attributes with observed human-robot motion could provide grounded distributions of clearance, yielding, speed, and gap acceptance for different deployment settings, allowing population robustness to be evaluated using observed rather than synthetic behavioral variation.

VII. LIMITATIONS

The behavioral populations in this study are synthetic and should be interpreted as controlled simulation inputs rather than estimates of particular cities, cultures, or demographic groups. The persona-to-motion pipeline is not calibrated using joint observations of persona attributes, encounter responses, and pedestrian motion from the same participants. The matched time-pressure intervention is similarly a controlled

perturbation of the synthetic records and should not be interpreted as an estimate of the causal effect of deadlines on real pedestrians. Alternative mappings, pedestrian dynamics models, and language models preserve the main qualitative pattern while changing reversal magnitudes and some controller orderings. The simulation also abstracts several aspects of real pedestrian–robot interaction. Pedestrians are represented as disk agents, assistive-device geometry and group behavior are not modeled, and the robot receives perfect local state information. The Human-like dynamics analysis is limited to a smaller obstacle-free subset. Future work can combine empirically measured behavioral populations with richer pedestrian models and real-world robot experiments.

VIII. CONCLUSION

This paper introduced PopNavShift to test the stability of social-navigation strategy evaluations across pedestrian population shifts. Across matched episodes, rankings based on pedestrian burden were more sensitive than those based on robot travel time, particularly in spatially constrained settings. Changing only time pressure for the same personas frequently reversed rankings based on mean and worst-decile pedestrian delay, and this pattern persisted across alternative mappings, pedestrian dynamics models, and language models. These results support evaluating social-navigation strategies across multiple behavioral populations and reporting pedestrian burden alongside robot-centered performance.

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